

ESASHI, Japan

WMO: 474280

Lat: 41.8672N

Lon: 140.1242E

Elev: 40

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.1	22.1	9.7	9.0	23.3	11.7	10.0	24.9	37.9	29.4	35.3	27.5	18.5	300	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.6	81.9	74.3	79.9	73.1	78.0	71.9	75.7	80.4	74.4	78.6	73.0	77.1	8.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.3	128.1	78.9	72.9	122.2	77.7	71.5	116.3	76.7	39.3	80.7	38.0	78.6	36.8	77.3	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
32.5	29.2	26.4	16.3	85.9	2.9	3.0	14.2	88.1	12.5	89.9	10.8	91.6	8.7	93.8
			14.4	76.9	2.5	2.1	12.6	78.5	11.1	79.7	9.7	81.0	7.8	82.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	30.4	31.4	36.8	45.6	54.2	61.6	69.5	73.2	67.2	56.3	45.0	34.9
	DBStd	15.47	4.69	5.31	5.28	4.88	4.44	4.06	3.92	3.79	4.85	5.24	6.67	5.66
	HDD50	2353	609	522	408	148	12	0	0	0	0	9	176	469
	HDD65	5763	1074	942	873	582	335	115	7	1	30	271	599	934
	CDD50	2577	0	0	0	16	142	349	604	718	515	206	27	0
	CDD65	512	0	0	0	0	0	14	146	254	96	2	0	0
	CDH74	1770	0	0	0	0	2	20	337	1127	284	0	0	0
	CDH80	173	0	0	0	0	0	1	28	115	29	0	0	0
(14) Wind	WSAvg	10.4	16.0	15.0	13.2	9.8	7.1	5.6	5.7	6.0	6.9	9.8	13.8	16.1
(15) Precipitation	PrecAvg	48.4	3.3	2.7	2.4	2.9	3.9	3.0	5.0	6.6	5.7	4.1	4.6	4.2
	PrecMax	61.9	5.4	6.4	3.9	5.2	8.9	7.0	14.6	16.9	10.4	7.2	8.6	6.8
	PrecMin	37.0	1.8	0.9	1.1	0.4	1.6	1.1	2.2	2.2	1.7	2.1	2.6	2.0
	PrecStd	6.2	1.0	1.1	0.8	1.3	1.7	1.5	2.5	3.4	2.4	1.3	1.5	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	48.4	53.3	63.9	71.0	77.0	83.1	85.4	83.7	71.6	63.5	53.7
		MCWB	40.7	44.2	46.9	53.1	58.9	66.4	73.5	76.6	74.3	63.9	57.8	49.0
	2%	DB	41.9	44.1	49.8	59.2	67.3	72.4	79.7	82.8	79.4	69.0	60.6	49.5
		MCWB	37.8	39.9	44.8	51.2	57.4	64.6	72.4	75.2	72.0	61.9	54.9	45.1
	5%	DB	39.7	41.5	47.4	55.8	64.2	70.1	77.4	81.0	77.1	67.0	58.2	46.4
		MCWB	35.4	37.3	43.0	49.2	56.5	63.4	71.3	74.2	70.3	60.2	52.8	41.8
	10%	DB	37.5	39.3	45.2	53.4	61.5	68.1	75.5	79.4	75.0	65.0	55.8	43.6
		MCWB	33.4	35.0	41.0	48.0	55.4	62.7	70.7	73.2	68.4	58.5	50.4	38.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.6	45.3	48.3	55.6	61.2	68.1	75.8	78.1	75.6	66.0	58.8	49.3
		MCDB	44.5	47.7	51.5	60.9	67.1	72.7	80.9	82.6	81.5	70.3	62.4	53.0
	2%	WB	38.2	40.4	45.8	52.6	59.3	66.3	74.1	76.6	73.6	63.1	55.7	45.6
		MCDB	41.4	43.6	49.4	57.7	64.6	70.1	77.5	81.4	78.5	67.4	59.7	49.2
	5%	WB	35.9	37.8	43.6	50.3	57.7	64.8	72.9	75.4	71.6	61.2	53.4	42.0
		MCDB	39.2	41.1	46.8	54.7	62.7	68.7	76.3	79.8	75.9	65.9	58.0	46.0
	10%	WB	33.7	35.4	41.4	48.6	56.2	63.4	71.5	74.2	69.3	59.3	50.8	39.1
		MCDB	37.2	38.9	45.0	52.8	60.8	67.4	75.0	78.5	74.1	64.5	55.4	43.1
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.2	9.5	10.9	11.0	10.0	8.8	9.6	11.5	11.9	10.0	8.3
		MCDBR	10.5	11.1	12.5	15.1	15.9	13.4	11.5	11.3	12.6	13.4	13.6	12.5
	5% WB	MCWBR	9.8	10.4	10.7	10.3	9.6	8.0	6.6	6.4	7.4	9.3	12.0	11.5
		MCWBR	10.5	11.3	11.9	13.1	13.2	10.5	9.3	9.4	10.5	12.0	13.4	12.6
(40) Clear-Sky Solar Irradiance	taub		0.365	0.412	0.484	0.540	0.556	0.526	0.523	0.484	0.427	0.414	0.403	0.384
		taud	2.077	1.947	1.850	1.786	1.830	1.987	2.069	2.193	2.305	2.191	2.114	2.056
	Ebn at Noon		234	240	237	235	235	243	242	248	255	240	219	213
		Edh at Noon	35	47	58	67	65	56	51	44	37	36	34	33
(44) All-Sky Solar Radiation	RadAvg		386	624	1042	1440	1600	1577	1386	1395	1204	882	494	335
	RadStd		28	65	82	146	151	151	132	124	115	60	49	25

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page